
title: “Nash Equilibria as Hopfield Energy Minima: A Field-Theoretic Foundation for Game Theory” subtitle: “Fractal Programme — Economic Scale (10–12)” author: “Alistair Johnson” orcid: “0009-0007-2194-0850” institute: “Independent Researcher, Zurich, Switzerland” date: “2026” lang: en-GB bibliography: ../../paper/bibliography.bib csl: ../../paper/apa-7th.csl abstract: |
We identify Nash equilibria in strategic-form games with energy minima of the Hopfield Hamiltonian, formally co-identifying classical game theory with the attractor dynamics of the Universal Somatic Field (USF) at the economic scale (scales 10–12, 10^3 – 10^4 m). The identification is exact: a Nash equilibrium profile $(s_1^*, \dots, s_N^*)$ is a fixed point of the best-response dynamics if and only if it is a local minimum of the energy function $H(s) = -\frac{1}{2}s^T W s$, where W_{ij} encodes the payoff externalities between players i and j . This is not an analogy; it is a structural identity whose proof is straightforward. We derive three consequences: (1) games with unique Nash equilibria correspond to Hopfield networks with isolated ground states; (2) coordination failures (multiple equilibria) correspond to the multi-attractor regime; (3) market crashes and economic crises are topological phase transitions — abrupt transitions between attractor basins triggered by a change in the spectral gap of W . The FM-HN extension of the USF further predicts that volitional intervention (policy, regulation, the “God-Knob”) can induce quantum tunnelling between equilibria, achieving coordination that classical best-response dynamics cannot reach.